

31 August 2026

Dear Editor,

We are pleased to resubmit our revised manuscript, “Wind Associations with Overwintering Monarch Butterfly Cluster Size Depend on Temperature and Sun Exposure,” for consideration in the *Insects* Special Issue “Advances in Monarch Butterfly Biology, Conservation, and Management: From Local to Continental.”

Wind protection has long been an important component of western monarch overwintering habitat management. In particular, winds at or above 2 m/s have been described as disruptive to monarch aggregations and have influenced grove restoration guidance in California.

We evaluated the predicted decline in cluster size by pairing time-lapse images with nearby wind measurements at two overwintering groves during one season. The observational design provided repeated measurements across 80 days while preserving the limitations of inference from two groves and one season.

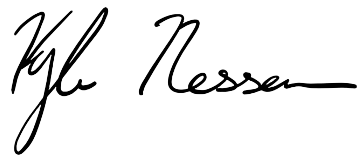
Stronger maximum gusts were not consistently followed by decreases in visible cluster size. Instead, the association depended on approximate local temperature and sun-exposed Butterfly Index. Under cool, sunny conditions, stronger gusts were associated with declines. Under warm, sunny conditions, they were associated with increases. These conditional patterns provide a basis for future tests of physiological and energetic mechanisms, but the present study did not measure those mechanisms directly.

The revised manuscript is shorter, limits its claims to the monitored conditions, corrects the model-comparison workflow, and makes the processed data and analytical materials public. The findings raise a focused question about applying a single wind threshold and identify clear priorities for replication across sites, seasons, grove structures, and cluster densities. We believe this combination of empirical monitoring, methodological transparency, and conservation relevance fits the scope of the special issue.

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal. All authors have approved the manuscript and agree with its submission to *Insects*.

Thank you for considering our work. We look forward to your response.

Sincerely,

A handwritten signature in black ink, reading "Kyle Nessen". The signature is fluid and cursive, with the first name "Kyle" and last name "Nessen" clearly legible.

Kyle Nessen

Corresponding author

On behalf of the co-authors Peter C. Ibsen, Jay E. Diffendorfer, and Francis X. Villablanca
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